Week 1: Robot Kinematics

Whiteboard Companion Notes

AI in Robotics

1 Configuration Space and DOF

1.1 Definitions

Configuration: Specification of the position of every point of a robot.

Degrees of Freedom (DOF): Minimum number n of real coordinates needed to represent configuration.

Configuration Space (C-space): Space of all possible configurations; configuration is a point in C-space.

General DOF Rule:

$$\text{dof} = \text{variables} - \text{independent constraints} \quad (1)$$

1.2 Rigid Body DOF

Planar rigid body: 3 DOF (2 position + 1 orientation)

Spatial rigid body: 6 DOF (3 position + 3 orientation)

1.3 Robot Joints

Joint	DOF	Planar c	Spatial c
Revolute (R)	1	2	5
Prismatic (P)	1	2	5
Helical (H)	1	—	5
Cylindrical (C)	2	—	4
Universal (U)	2	—	4
Spherical (S)	3	—	3

Table 1: Joint types: DOF f_i and constraints c_i

1.4 Grübler's Formula

For a mechanism with N links (including ground), J joints, and $m = 3$ (planar) or $m = 6$ (spatial):

$$\text{dof} = m(N - 1 - J) + \sum_{i=1}^{J} f_i \quad (2)$$

where f_i is the DOF of joint i.

Examples:

- **Four-bar linkage:** $N = 4$, $J = 4$ (all R), $\text{dof} = 3(4 - 1 - 4) + 4 = 1$
- **k-link serial chain:** $N = k+1$, $J = k$, $\text{dof} = k$
- **Stewart-Gough platform:** $N = 14$, $J = 18$ (6U+6P+6S), $\text{dof} = 6$

1.5 C-Space Topology

Common topological spaces:

- S^1 = circle (revolute joint)
- $\mathbb{R}^1$ = line (prismatic joint)
- $T^n = S^1 \times \cdots \times S^1 = n$-torus

Examples:

- Planar rigid body: $\mathbb{R}^2 \times S^1$
- 2R robot: T^2
- PR robot: $\mathbb{R}^1 \times S^1$
- Spatial rigid body: $\mathbb{R}^3 \times S^2 \times S^1$

1.6 Constraints

Holonomic (integrable): $g(\theta) = 0$ reduces C-space dimension.

Nonholonomic (Pfaffian): $A(\theta)\dot{\theta} = 0$ restricts velocities but not reachable configurations.

Example: Rolling coin no-slip constraint is nonholonomic.

1.7 Task Space vs Workspace

Task space: Space in which the task is naturally expressed (e.g., $\mathbb{R}^3$ for position, $\mathbb{R}^3 \times S^2$ for spray painting).

Workspace: Set of end-effector configurations reachable by the robot.

2 Rigid-Body Transformations

2.1 Rotation Matrices: SO(3)

The special orthogonal group:

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times 3} : R^T R = I, \det R = 1\} \tag{3}$$

Properties:

- $R^{-1} = R^T$
- $R_1 R_2 \in \mathrm{SO}(3)$ (closed under composition)
- Preserves distances: $\|Rx\| = \|x\|$

Elementary rotations:

$$\mathrm{Rot}(\hat{x}, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\theta & -s_\theta \\ 0 & s_\theta & c_\theta \end{bmatrix} \tag{4}$$

$$\mathrm{Rot}(\hat{y}, \theta) = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \tag{5}$$

$$\mathrm{Rot}(\hat{z}, \theta) = \begin{bmatrix} c_\theta & -s_\theta & 0 \\ s_\theta & c_\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{6}$$

2.2 Angular Velocity and so(3)

Skew-symmetric matrix: For $\omega = [\omega_1, \omega_2, \omega_3]^T$,

$$[\omega] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \in \mathfrak{so}(3) \tag{7}$$

Properties:

- $[\omega]x = \omega \times x$
- $[\omega] = -[\omega]^T$
- $R[\omega]R^T = [R\omega]$

Velocity relation:

$$\dot{R} = [\omega_s]R \quad \text{(spatial frame)} \tag{8}$$

$$\dot{R} = R[\omega_b] \quad \text{(body frame)} \tag{9}$$

2.3 Exponential Map: Rodrigues Formula

For unit axis $\hat{\omega}$ and angle θ:

$$e^{[\hat{\omega}]\theta} = I + \sin\theta\,[\hat{\omega}] + (1-\cos\theta)\,[\hat{\omega}]^2 \tag{10}$$

This maps $\mathfrak{so}(3) \to \mathrm{SO}(3)$.

2.4 Matrix Logarithm

Given $R \in \mathrm{SO}(3)$, find $[\hat{\omega}]\theta$:

If $\sin\theta \neq 0$:

$$\theta = \cos^{-1}\left(\frac{\mathrm{tr}(R) - 1}{2}\right) \tag{11}$$

$$[\hat{\omega}] = \frac{1}{2\sin\theta}(R - R^T) \tag{12}$$

If $\theta = \pi$: $R = I + 2[\hat{\omega}]^2$, solve for $\hat{\omega}$.
If $\theta = 0$: $\hat{\omega}$ is undefined, $R = I$.

2.5 Homogeneous Transformations: SE(3)

The special Euclidean group:

$$\mathrm{SE}(3) = \left\{ T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} : R \in \mathrm{SO}(3), p \in \mathbb{R}^3 \right\} \tag{13}$$

Inverse:

$$T^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix} \tag{14}$$

Composition:

$$T_1 T_2 = \begin{bmatrix} R_1 R_2 & R_1 p_2 + p_1 \\ 0 & 1 \end{bmatrix} \tag{15}$$

2.6 Twists and se(3)

Twist: 6D velocity vector $V = \begin{bmatrix} \omega \\ v \end{bmatrix} \in \mathbb{R}^6$

Matrix form:

$$[V] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} \in \mathfrak{se}(3) \tag{16}$$

Body twist: $T^{-1}\dot{T} = [V_b]$, where $v_b = R^T \dot{p}$
Spatial twist: $\dot{T}T^{-1} = [V_s]$, where $v_s = \dot{p} - \omega_s \times p$

2.7 Screw Axes

A screw axis $S = \begin{bmatrix} \omega \\ v \end{bmatrix}$ satisfies either:

- $\|\omega\| = 1$ and $v = -\omega \times q + h\omega$ (finite pitch h, point q on axis)
- $\omega = 0$ and $\|v\| = 1$ (infinite pitch, pure translation)

Chasles-Mozzi theorem: Every rigid-body displacement is a screw motion.

2.8 Exponential Coordinates for SE(3)

For $\|\omega\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega]\theta} & G(\theta)v \\ 0 & 1 \end{bmatrix} \quad (17)$$

where

$$G(\theta) = I\theta + (1 - \cos\theta)[\omega] + (\theta - \sin\theta)[\omega]^2 \quad (18)$$

For $\omega = 0$, $\|v\| = 1$:

$$e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix} \quad (19)$$

2.9 Adjoint Representation

$$[\mathrm{Ad}_T] = \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \in \mathbb{R}^{6\times 6} \quad (20)$$

Change of frame for twists:

$$V_s = [\mathrm{Ad}_{T_{sb}}]V_b \quad (21)$$

Change of frame for wrenches:

$$F_b = [\mathrm{Ad}_{T_{ab}}]^T F_a \quad (22)$$

2.10 Alternative Rotation Representations

Euler angles: Three successive rotations (e.g., ZYX). *Suffers from gimbal lock.*

Quaternions: $q = (q_0, q_1, q_2, q_3)$ with $\|q\| = 1$. No gimbal lock, efficient for interpolation.

3 Forward Kinematics

3.1 Product of Exponentials (PoE)

Space form:

$$T(\theta) = e^{[S_1]\theta_1} e^{[S_2]\theta_2} \cdots e^{[S_n]\theta_n} M \quad (23)$$

where S_i is the screw axis of joint i in the base frame at zero position, and M is the end-effector frame at zero position.

Body form:

$$T(\theta) = M\, e^{[B_1]\theta_1} e^{[B_2]\theta_2} \cdots e^{[B_n]\theta_n} \quad (24)$$

where $B_i = [\mathrm{Ad}_{M^{-1}}]S_i$.

3.2 Screw Axes for Common Joints

Revolute joint: Axis $\hat{\omega}$, point q on axis:

$$S = \begin{bmatrix} \hat{\omega} \\ -\hat{\omega} \times q \end{bmatrix} \quad (25)$$

Prismatic joint: Direction $\hat{v}$:

$$S = \begin{bmatrix} 0 \\ \hat{v} \end{bmatrix} \quad (26)$$

3.3 Example: 3R Spatial Robot

Suppose joint 1 is Z-axis at origin, joint 2 is Y-axis at $(0, 0, L_1)$, joint 3 is X-axis at $(0, L_2, L_1)$:

$$S_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad S_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ -L_1 \end{bmatrix}, \quad S_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -L_2 \\ 0 \end{bmatrix} \quad (27)$$

3.4 URDF Basics

Universal Robot Description Format: XML file describing:

- `<link>`: mass, inertia, geometry
- `<joint>`: type (revolute, prismatic, fixed), parent/child links, axis, origin

Tree structure; closed loops need additional constraints.

4 Jacobian and Velocity Kinematics

4.1 Space Jacobian

$$V_s = J_s(\theta)\dot{\theta} \quad (28)$$

Columns:

$$J_{si}(\theta) = \begin{cases} S_1 & i = 1 \\ [\mathrm{Ad}_{e^{[S_1]\theta_1} \cdots e^{[S_{i-1}]\theta_{i-1}}}](S_i) & i > 1 \end{cases} \quad (29)$$

4.2 Body Jacobian

$$V_b = J_b(\theta)\dot{\theta} \quad (30)$$

Columns:

$$J_{bi}(\theta) = \begin{cases} [\mathrm{Ad}_{e^{-[B_n]\theta_n} \cdots e^{-[B_{i+1}]\theta_{i+1}}}](B_i) & i < n \\ B_n & i = n \end{cases} \quad (31)$$

4.3 Jacobian Relationship

$$J_s(\theta) = [\mathrm{Ad}_{T_{sb}(\theta)}]J_b(\theta) \tag{32}$$

4.4 Statics

Power balance $\tau^T\dot{\theta} = F^T V$ gives:

$$\tau = J^T(\theta)F \tag{33}$$

If J is square and invertible:

$$F = J^{-T}\tau \tag{34}$$

4.5 Singularities

Definition: Configuration where $\mathrm{rank}(J) < \min(6, n)$.

At singularities:

- End-effector loses instantaneous DOF
- Robot can support arbitrary wrenches in certain directions

Common singularities for 6R chains:

- Collinear revolute axes
- Three coplanar parallel axes
- Four axes through one point (wrist)

4.6 Manipulability Ellipsoid

For unit joint velocity $\|\dot{\theta}\| = 1$:

$$\dot{q}^T(JJ^T)^{-1}\dot{q} = 1 \tag{35}$$

Ellipsoid axes: eigenvectors of JJ^T; lengths: $\sqrt{\lambda_i}$.

Manipulability measure:

$$\mu = \sqrt{\det(JJ^T)} \tag{36}$$

Zero at singularities.

5 Inverse Kinematics

5.1 Analytic IK: 2R Planar

For target (x, y) with link lengths L_1, L_2:

Law of cosines:

$$\cos\theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} \tag{37}$$

Two solutions (elbow-up/down):

$$\theta_2 = \pm\cos^{-1}\left(\frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}\right) \tag{38}$$

$$\theta_1 = \mathrm{atan2}(y, x) - \mathrm{atan2}(L_2\sin\theta_2, L_1 + L_2\cos\theta_2) \tag{39}$$

No solution if: $x^2 + y^2 > (L_1 + L_2)^2$ or $< |L_1 - L_2|^2$.

5.2 Analytic IK: 6R PUMA-Type

Strategy: Decouple position (first 3 joints) and orientation (wrist).

1. Compute wrist center p_w from desired T_{sd}
2. Solve for $\theta_1, \theta_2, \theta_3$ (shoulder + elbow) using geometry
3. Solve for $\theta_4, \theta_5, \theta_6$ (wrist) from orientation

Typically 4 position solutions (lefty/righty $\times$ elbow-up/down) and 2 wrist solutions, giving up to 8 IK solutions.

5.3 Numerical IK: Newton-Raphson

Goal: Given T_{sd}, find θ such that $T(\theta) = T_{sd}$.

Algorithm (body frame):

1. Initialize: θ_0, $i = 0$
2. Compute: $[V_b] = \log(T_{sb}^{-1}(\theta_i)T_{sd})$
3. While $\|\omega_b\| > \varepsilon_\omega$ or $\|v_b\| > \varepsilon_v$:
   - $\theta_{i+1} = \theta_i + J_b^\dagger(\theta_i)V_b$
   - $i \leftarrow i + 1$

Pseudoinverse: For redundant or deficient Jacobians, use Moore-Penrose pseudoinverse:

- Fat J ($n > m$): $J^\dagger = J^T(JJ^T)^{-1}$
- Tall J ($n < m$): $J^\dagger = (J^TJ)^{-1}J^T$

5.4 Inverse Velocity Kinematics

Given desired twist V_d:

$$\dot{\theta} = J^\dagger V_d \tag{40}$$

For redundant robots ($n > 6$), this gives minimum-norm solution:

$$\dot{\theta} = \arg\min_{\dot{\theta}} \|\dot{\theta}\|^2 \quad \text{s.t.} \quad J\dot{\theta} = V_d \tag{41}$$

Appendix: Formula Summary

$$
\begin{aligned}
\text{DOF} &= m(N-1-J) + \sum f_i \\
e^{[\hat{\omega}]\theta} &= I + \sin\theta\,[\hat{\omega}] + (1-\cos\theta)\,[\hat{\omega}]^2 \\
[\text{Ad}_T] &= \begin{bmatrix} R & 0 \\ [p]R & R \end{bmatrix} \\
T(\theta) &= e^{[S_1]\theta_1} \cdots e^{[S_n]\theta_n} M \\
V_s &= J_s(\theta)\dot{\theta}, \quad V_b = J_b(\theta)\dot{\theta} \\
J_s &= [\text{Ad}_{T_{sb}}] J_b \\
\tau &= J^T F \\
\theta_{i+1} &= \theta_i + J^{\dagger} V_b \quad \text{(IK iteration)}
\end{aligned}
$$